Interaction

Arpit Narechania

Agenda

- Define interaction - what is it?
- Talk about taxonomies for interaction
- Discuss examples for types of interaction

Background

- “The communication between user and the system” [Dix et al., 1998]
- “Direct manipulation and instantaneous change” [Becker et al., 1987]
- “HCI research is far from having solid (and falsifiable) theories of interaction” [Beaudouin-Lafon, 2004]

Interaction is often “Little Sibling” of InfoVis

- Two main components in an InfoVis
  - Representation
  - Interaction

- Representation gets all the attention…

- Interaction is often where the action is!
  - (no pun intended)

Research Focus

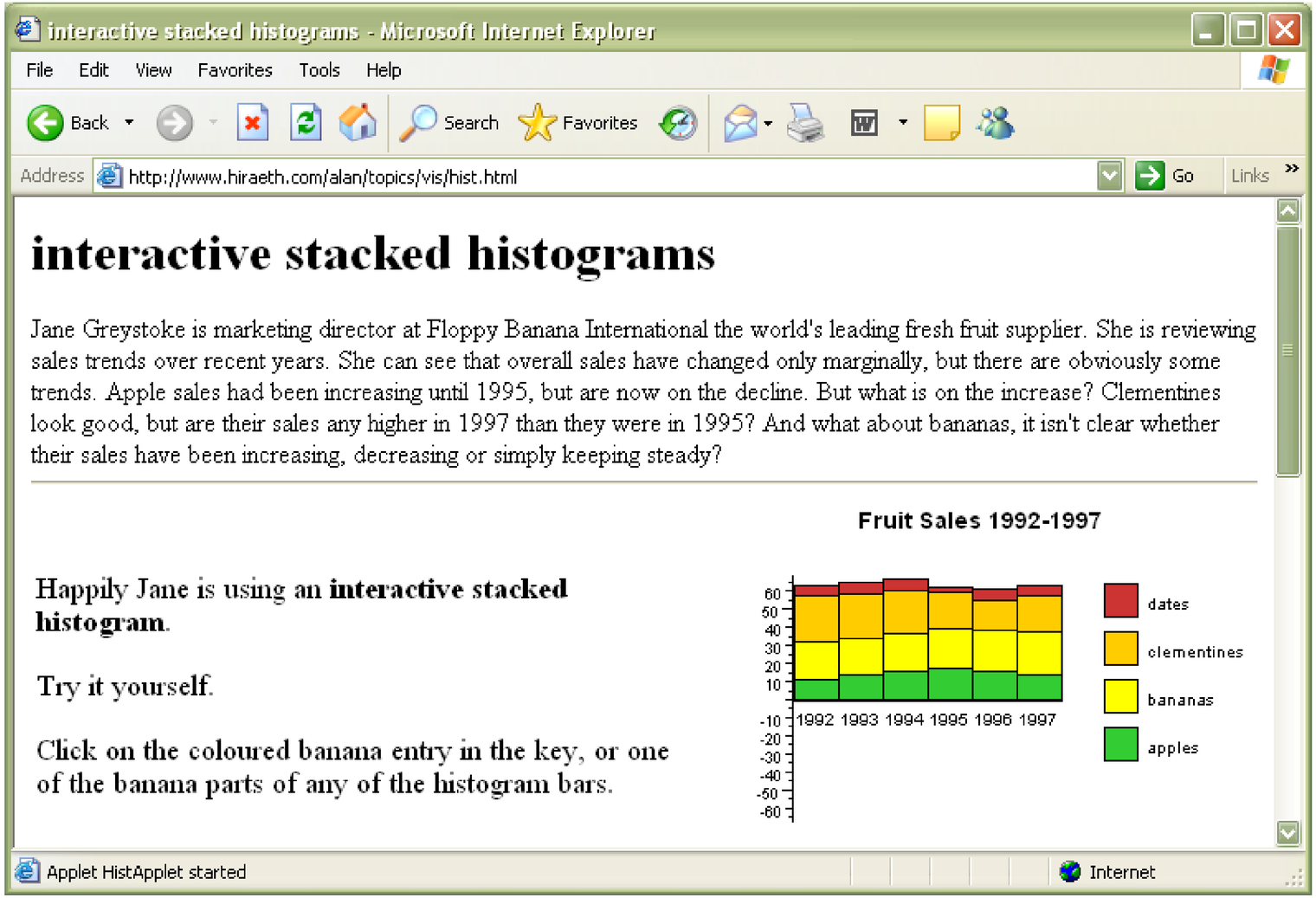
- Very challenging to come up with innovative, new visual representations now

- But can do interesting work on how users interact with the view(s)
  - It is what distinguishes infovis from static visual representations on paper.

Example

Stacked Histogram

Even simple interaction can be quite powerful



http://www.hiraeth.com/alan/topics/vis/hist.html

Taxonomies of User Interaction Types

Interaction Types

- Dix and Ellis (AVI ’98) propose
  - Highlighting and focus
  - Accessing extra info – drill down and hyperlinks
  - Overview and context – zooming and fisheyes
  - Same representation, changing parameters
  - Linking representations – temporal fusion

Interaction Types

- Keim's taxonomy (TVCG '02) includes
  - Projection
  - Filtering
  - Zooming
  - Distortion
  - Linking and brushing

Few’s Principles

- Comparing
- Sorting
- Adding variables
- Filtering
- Highlighting
- Aggregating
- Re-expressing
- Re-visualizing
- Zooming and panning
- Re-scaling
- Accessing details on demand
- Annotating
- Bookmarking

Now You See It Chapter 4

Another take

Toward a Deeper Understanding of the Role of Interaction in Information Visualization

Ji Soo Yi, Youn ah Kang, John T. Stasko, *Member, IEEE*, and Julie A. Jacko

Abstract—Even though interaction is an important part of information visualization (Infovis), it has garnered a relatively low level of attention from the Infovis community. A few frameworks and taxonomies of Infovis interaction techniques exist, but they typically focus on low-level operations and do not address the variety of benefits interaction provides. After conducting an extensive review of Infovis systems and their interactive capabilities, we propose seven general categories of interaction techniques widely used in Infovis: 1) Select, 2) Explore, 3) Reconfigure, 4) Encode, 5) Abstract/Elaborate, 6) Filter, and 7) Connect. These categories are organized around a user's intent while interacting with a system rather than the low-level interaction techniques provided by a system. The categories can act as a framework to help discuss and evaluate interaction techniques and hopefully lay an initial foundation toward a deeper understanding and a science of interaction.

Index Terms—Information visualization, interaction, interaction techniques, taxonomy, visual analytics

1 Introduction

Information visualization (Infovis) systems, at their core, appear to have two main components: representation and interaction. The representation component, whose roots lie in the field of computer graphics, concerns the mapping from data to representation and how that representation is rendered on the display. The interaction component involves the dialog between the user and the system as the user explores the data set to uncover insights. The interaction component's roots lie in the area of human-computer interaction (HCI). Although discussed as two separate components, representation and interaction clearly are not mutually exclusive. For instance, interaction with a system may activate a change in representation. Nonetheless, the two components seem to compose the two fundamental aspects of Infovis systems, and it seems reasonable to consider what each contributes to an end-user's and jotting down notes on the poster). Spence even suggests the notion of "passive interaction" through which the user's mental model on the data set is changed or enhanced [38]. Finally, through interaction, some limits of a representation can be overcome, and the cognition of a user can be further amplified (e.g., [15, 29]).

The importance of interaction and the need for its further study seem undisputed. For example, the recent book *Illuminating the Path: The Research and Development Agenda for Visual Analytics* calls for further research on interaction:

> "**Recommendation 3.3: Create a new science of interaction to support visual analytics.** The grand challenge of interaction is to develop a taxonomy to describe the design space of interaction techniques that supports the

IEEE TVCG 2007

Study Methodology

- Survey
  - 59 papers
    - Papers introducing new interaction systems
    - Well-known papers in subareas of Infovis
  - 51 systems
    - Commercial Infovis Systems (SeeIT, Spotfire, TableLens, InfoZoom, etc.)
  - Collected 311 individual interaction techniques

- Affinity Diagram Method

Results

- 7 categories
  - Select
  - Explore
  - Reconfigure
  - Encode
  - Abstract/Elaborate
  - Filter
  - Connect

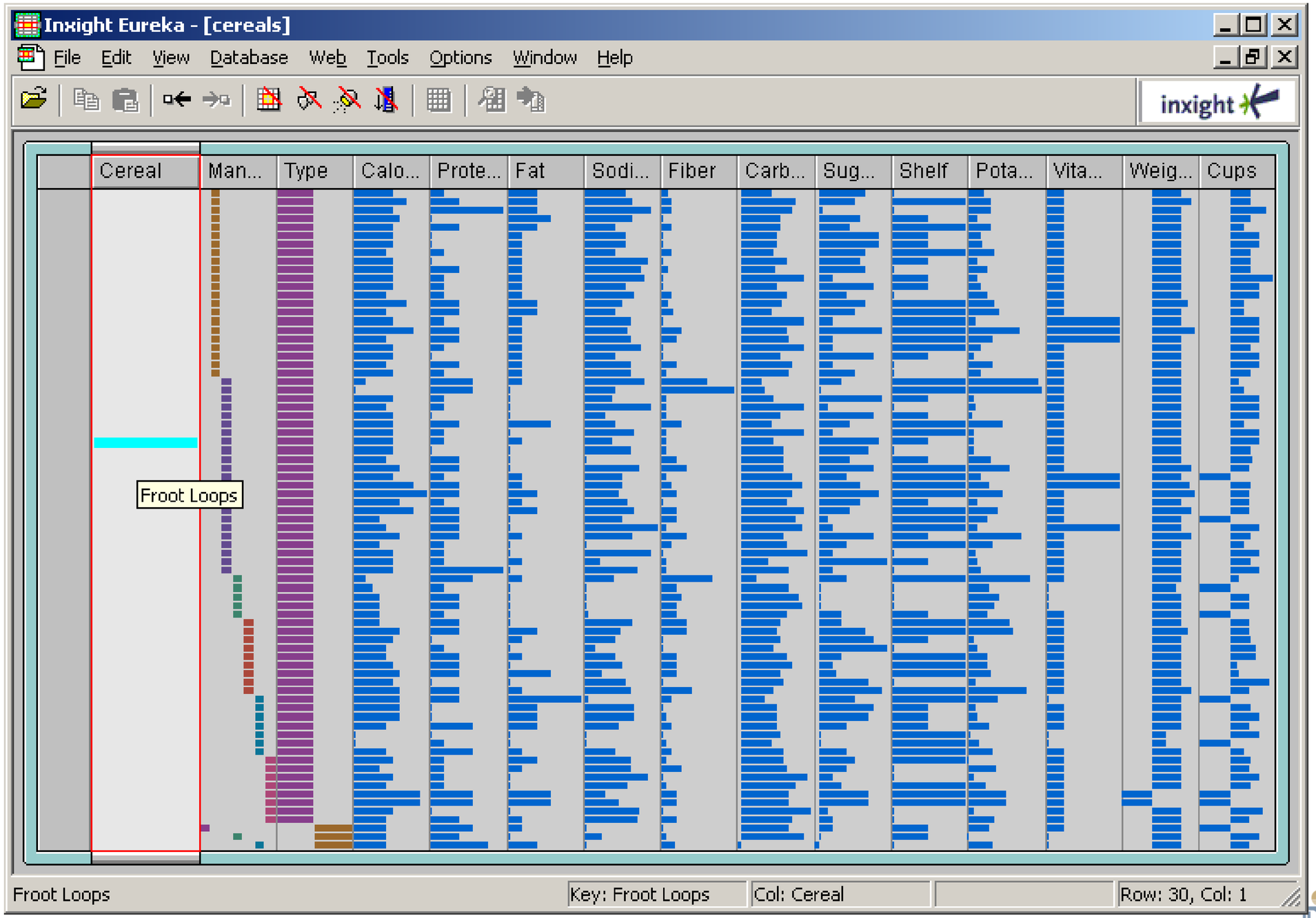
1. Select

- “Mark something as interesting”
  - Mark items of interest to keep track
  - Often as a preceding action to subsequent operations.

- e.g.,
  - Selecting a placemark in Google Maps
  - The Focus feature in TableLens

Pop-up tooltips

Hovering mouse cursor shows item details

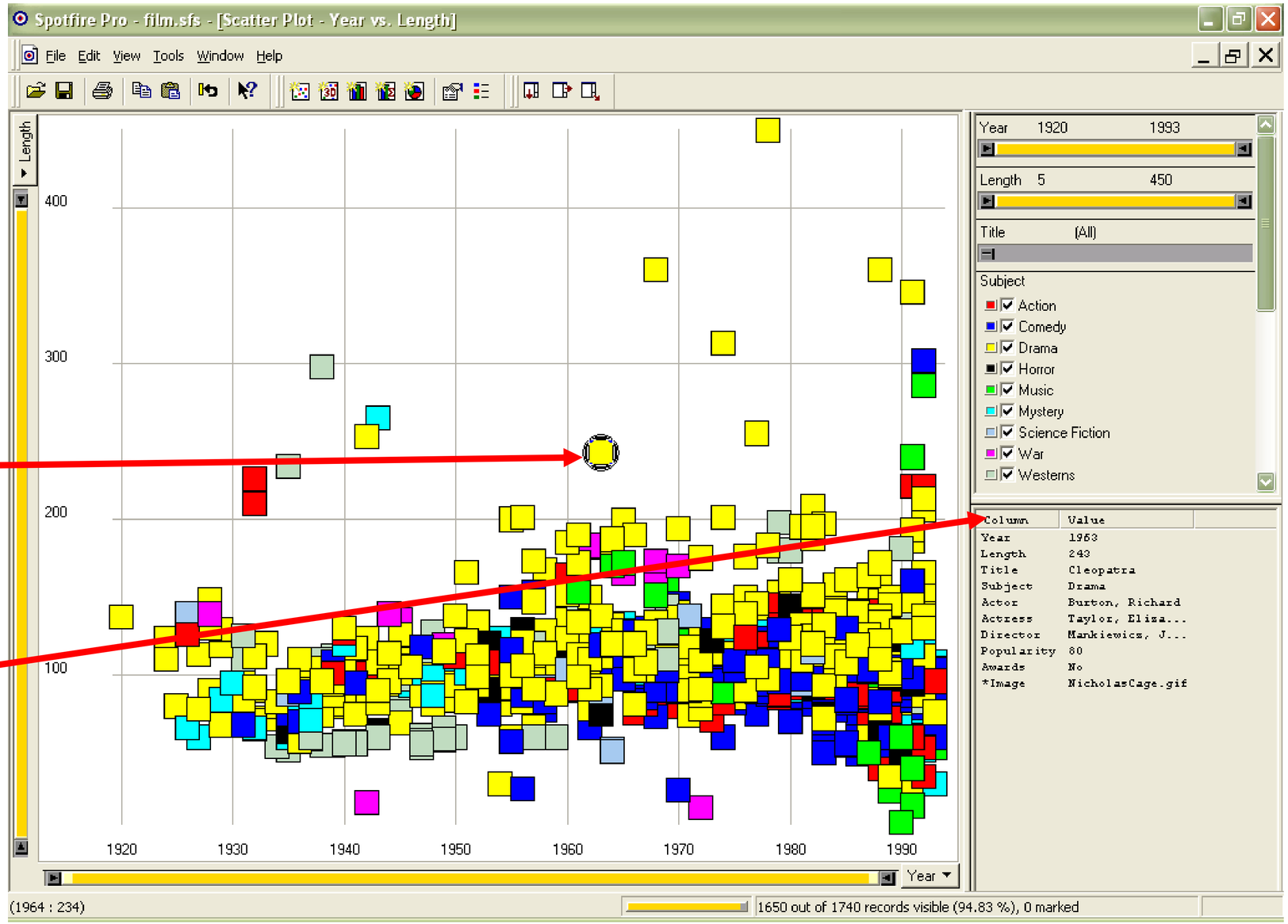


Mouse Selection

Clicking on an item selects it and attributes of the data point are shown

Selected item

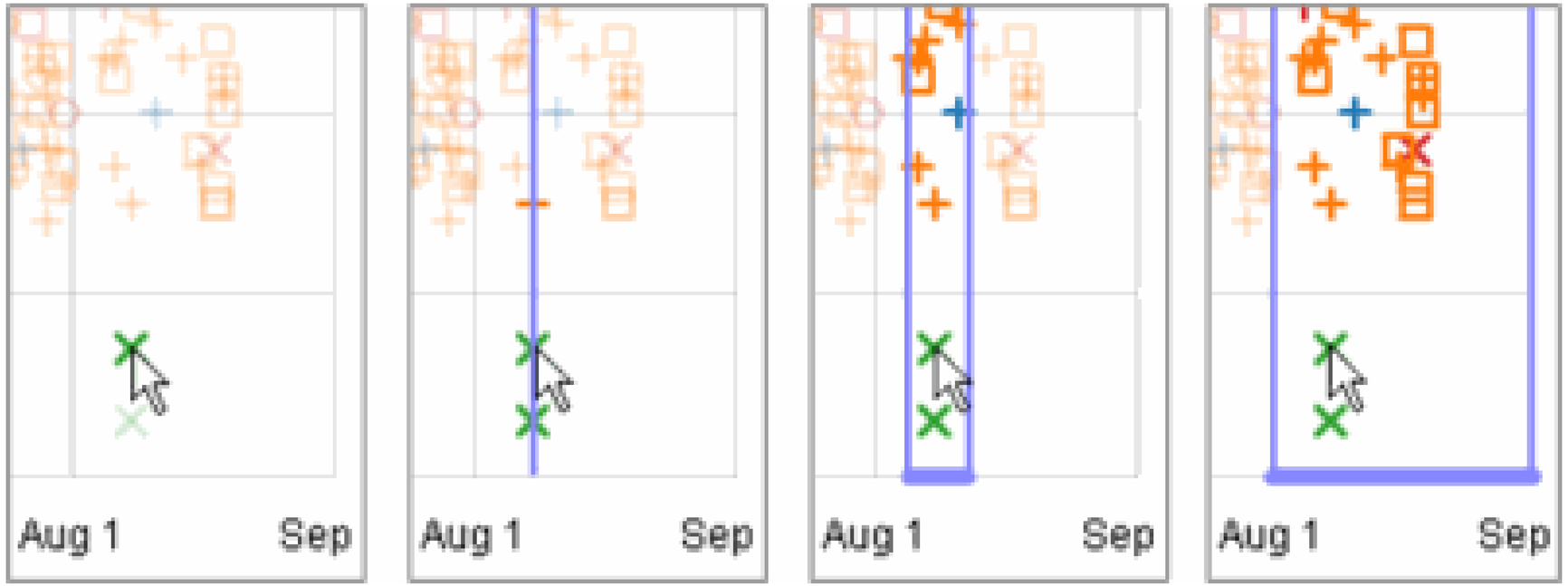
Attributes



Generalized Selection

- When you click on an item in a visualization, can we generalize the selection from the precise item?
  - Maybe you want to select items matching some attribute(s) of that item
  - Maybe you wanted items that are nearby your selection?

Query Relaxation



As you dwell on your mouse click, the selection criteria broaden and you can choose sets of items

Video at: http://vis.berkeley.edu/papers/generalized_selection/ and https://youtu.be/LVMjfsoMZfg?t=101

Heer, Agrawala, Willett
CHI '08

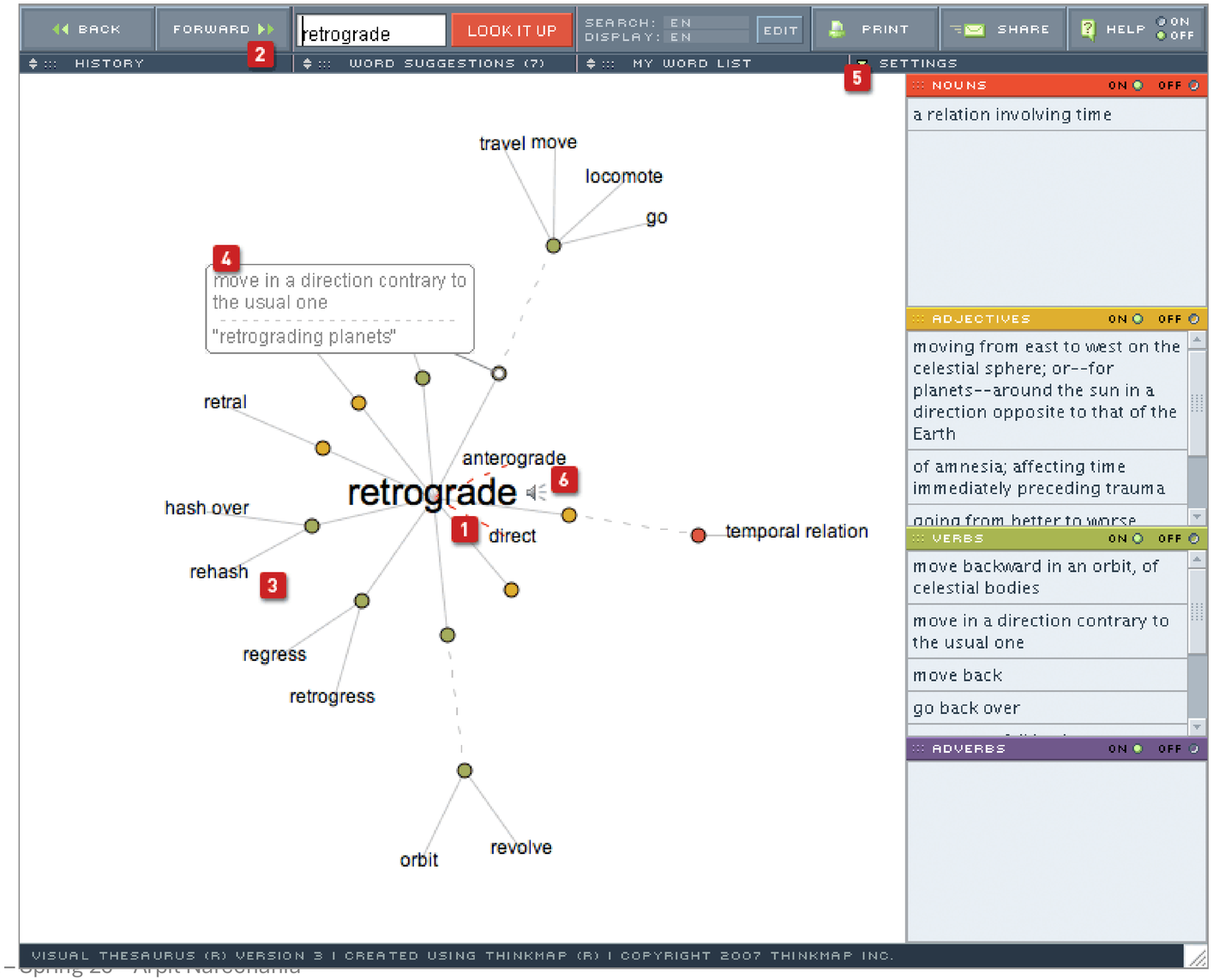
2. Explore

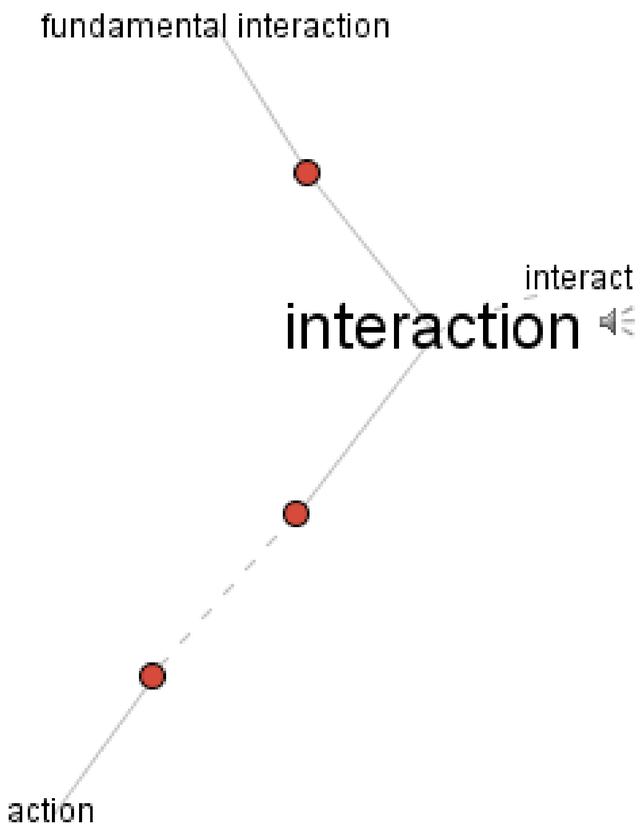
- “Show me something different”
  - Enable users to examine a different subset of data
  - Overcome the limitation of display size

- e.g.,
  - Panning in Google Earth
  - Direct Walking in graphs (e.g., Visual Thesaurus)

Example

Visual Thesaurus: http://www.visualthesaurus.com





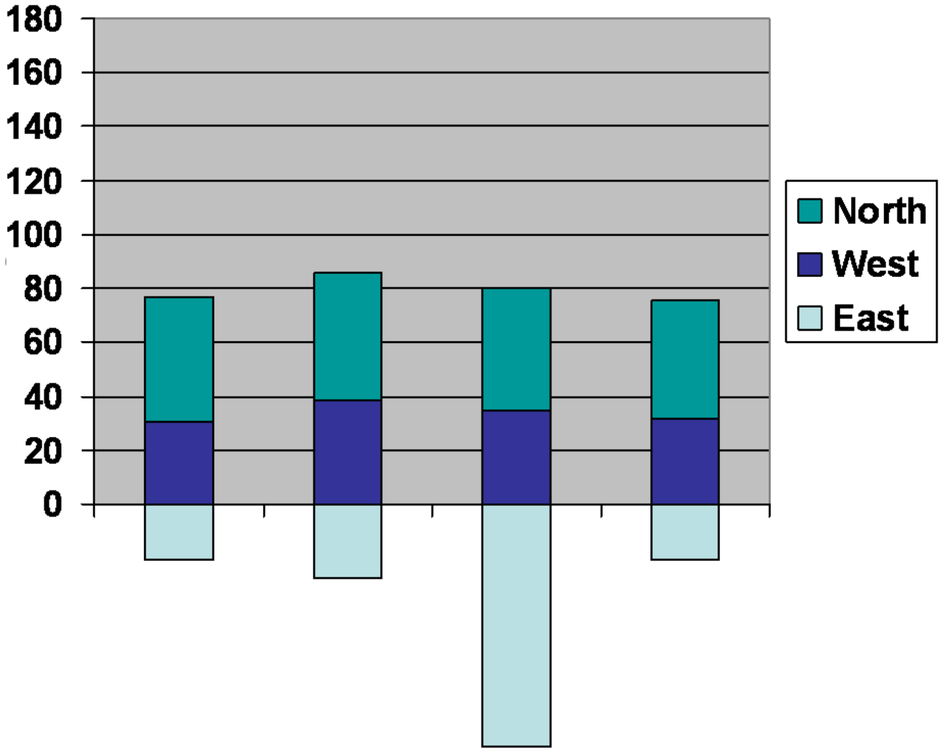
much more on this when we talk about graphs later

3. Reconfigure

- “Show me a different arrangement”
  - Provide different perspectives by changing the spatial arrangement of representation

- e.g.,
  - The baseline adjustment feature in Stacked Histogram
  - Sorting and rearranging columns in TableLens
  - Changing the attributes in a scatter plot
  - The “Spread Dust” feature in Dust & Magnet

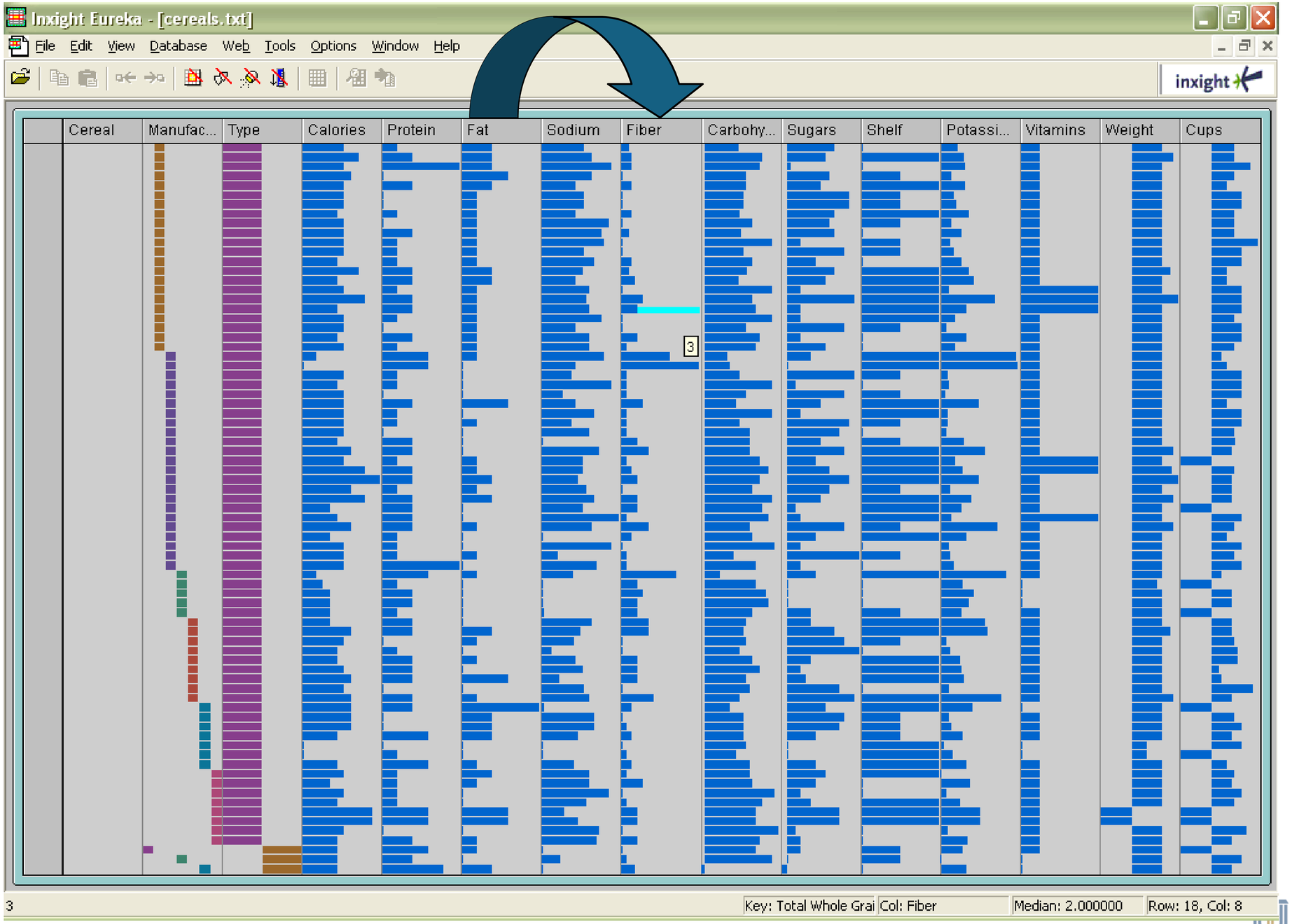
Example



Stacked Histogram

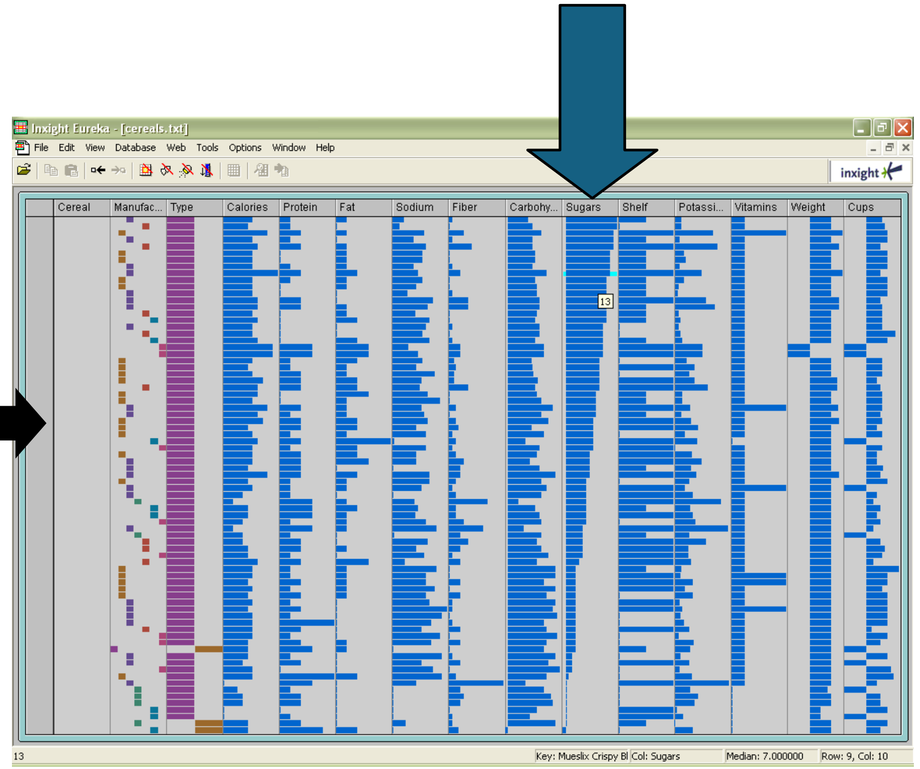
Rearrange

In TableLens you can move columns (attributes) left and right



Sorting

Can sort data with respect to a particular attribute in Table Lens

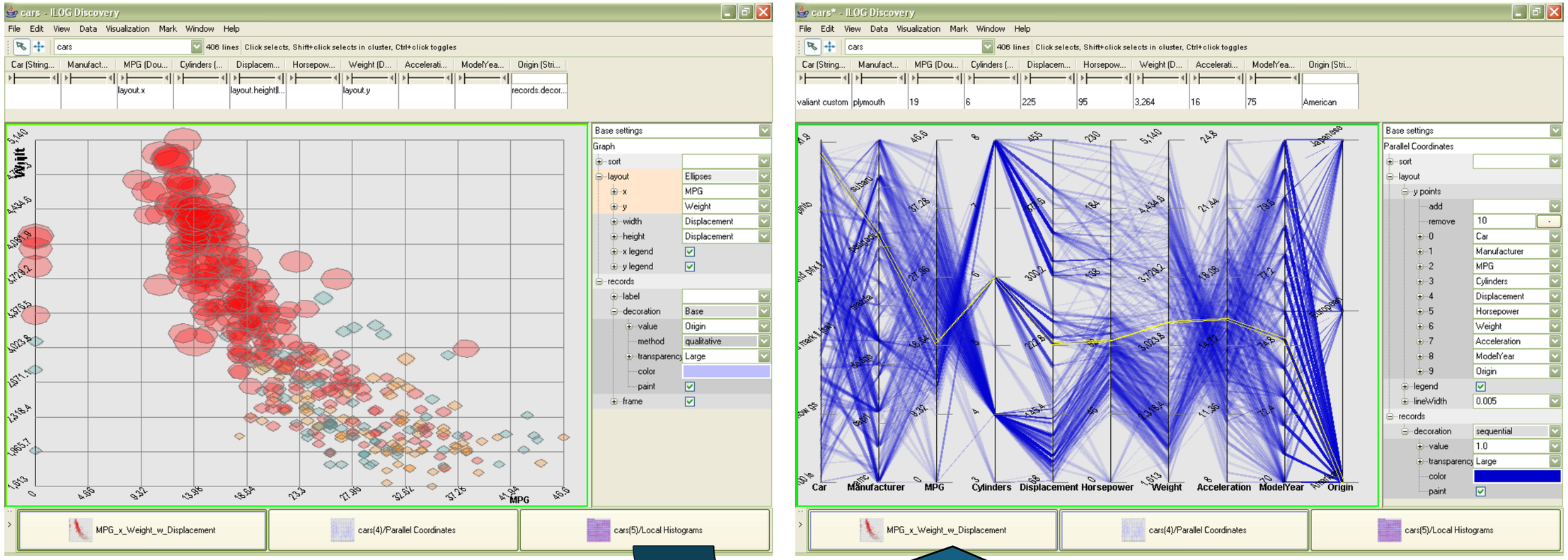


4. Encode

- “Show me a different representation”
  - Change visual appearances / encoding

- e.g.,
  - Changing color encoding
  - Changing size
  - Changing orientation
  - Changing font
  - Changing shape

Example



Switching views

5. Abstract/Elaborate

- “Show me more or less detail”
  - Adjust the level of abstraction (overview and details)

- e.g.,
  - Unfolding sub-categories in an interactive pie chart
  - Details-on-demand in Sunburst
  - The tool-tip operation in TableLens
  - Zooming (geometric zooming)

Details-on-Demand

- Term used in infovis when providing viewer with more information/details about data case or cases

- May just be more info about a specific data item (via selection)

- May be moving from aggregation view to individual view
  - May not be showing all the data due to scale problem
  - May be showing some abstraction of groups of elements
  - Expand set of data to show more details, perhaps individual cases

Examples

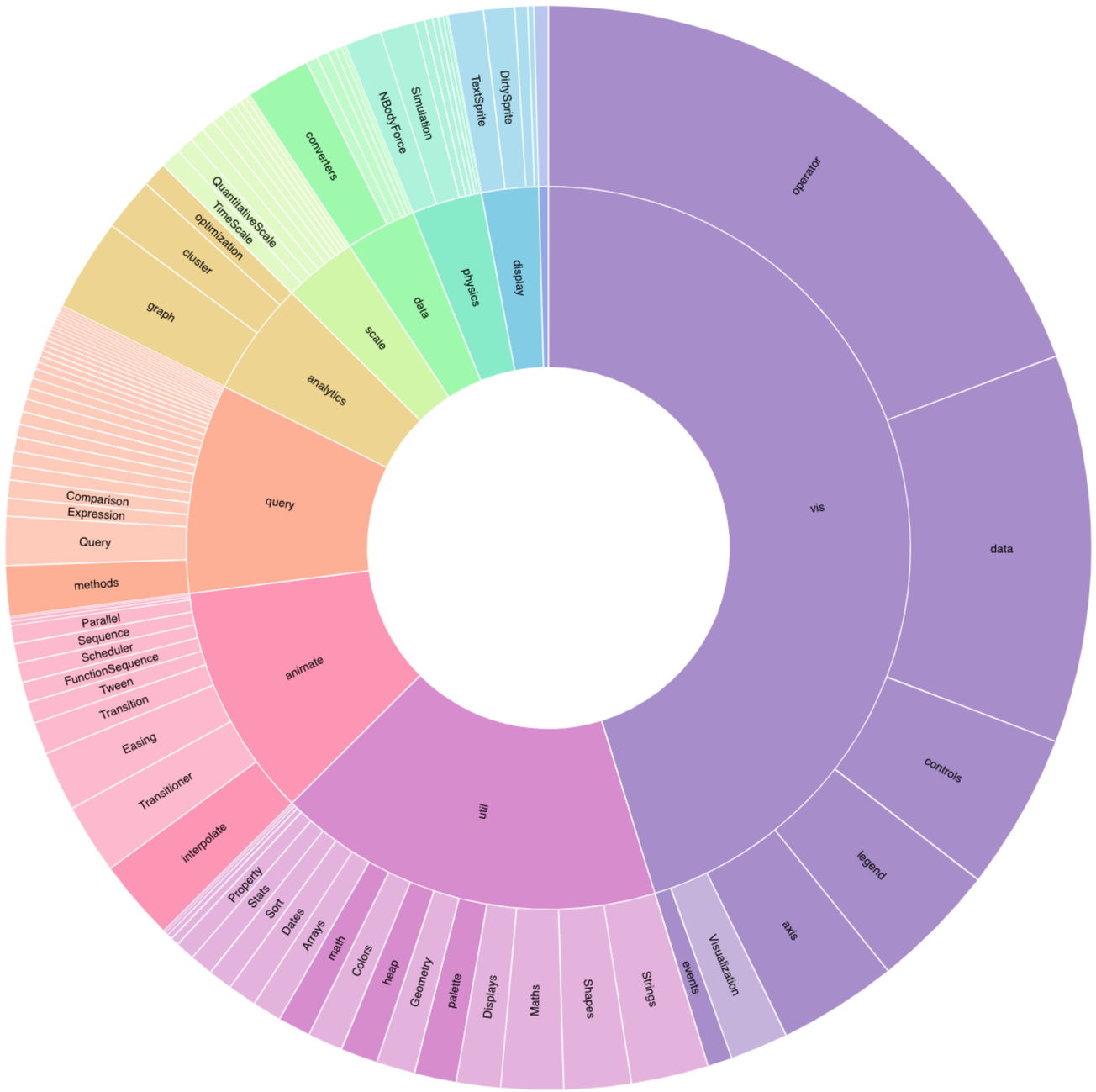
Table Lens

Example

https://observablehq.com/@d3/zoomable-sunburst

Zoomable Sunburst

This variant of a sunburst diagram shows only two layers of the hierarchy at a time. Click a node to zoom in, or the center to zoom out. Compare to an icicle.



6. Filter

- “Show me something conditionally”
  - Change the set of data items being presented based on some specific conditions.

- e.g.,
  - Dynamic query
  - Attribute Explorer
  - Keystoke based filtering in NameVoyager
  - QuerySketch

Dynamic Query

- Probably best-known and one of most useful infovis techniques
- Let’s explore more details…

DB Queries

- Query language
  - Select house-address
    From hk-realty-db
    Where price >= 2,000,000 and
    price <= 4,000,000 and
    bathrooms >= 5 and
    garage == 2 and
    bedrooms >= 5

DB Queries

Pros

- powerful, flexible

Cons

- one at a time
- difficult to construct

Typical Query Response

- 124 hits found
  1. Acacia, 150 Kennedy Road, Wan Chai – a beautiful …
  2. Ocean View, 89 Repulse Bay Road, Southern District – …
  3. …

- 0 hits found
  - great…. :/

Further Cons

- Must learn language/syntax
- Only shows exact matches
- Don’t know magnitude of results
- No helpful context is shown
- Reformulating to a new query can be slow
- …

Dynamic Query

- Specifying a query brings immediate display of results
- Responsive interaction (< .1 sec) with data, concurrent presentation of solution
- “Fly through the data”, promote exploration, make it a much more “live” experience

Dynamic Query Constituents

- Rapid, incremental and reversible actions
- Selection by pointing (not typing)
- Immediate and continuous display of results

Shneiderman
IEEE Software ’94

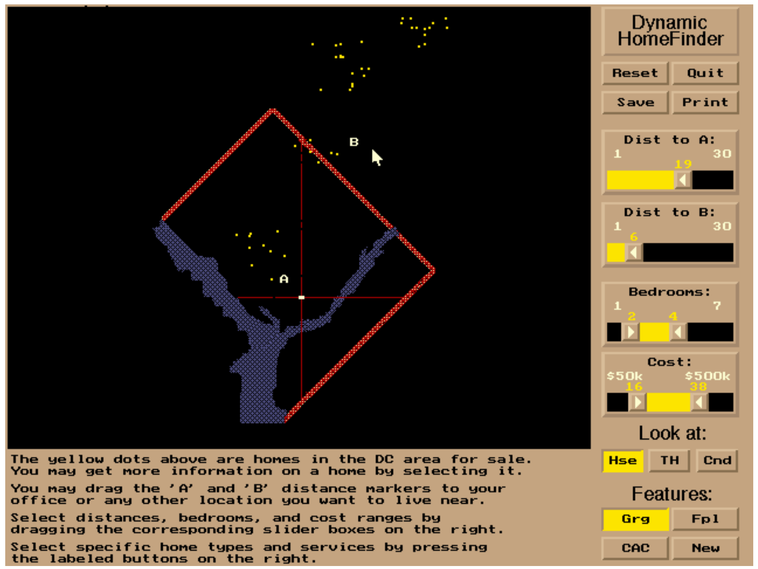
Ahlberg & Shneiderman
CHI ‘94

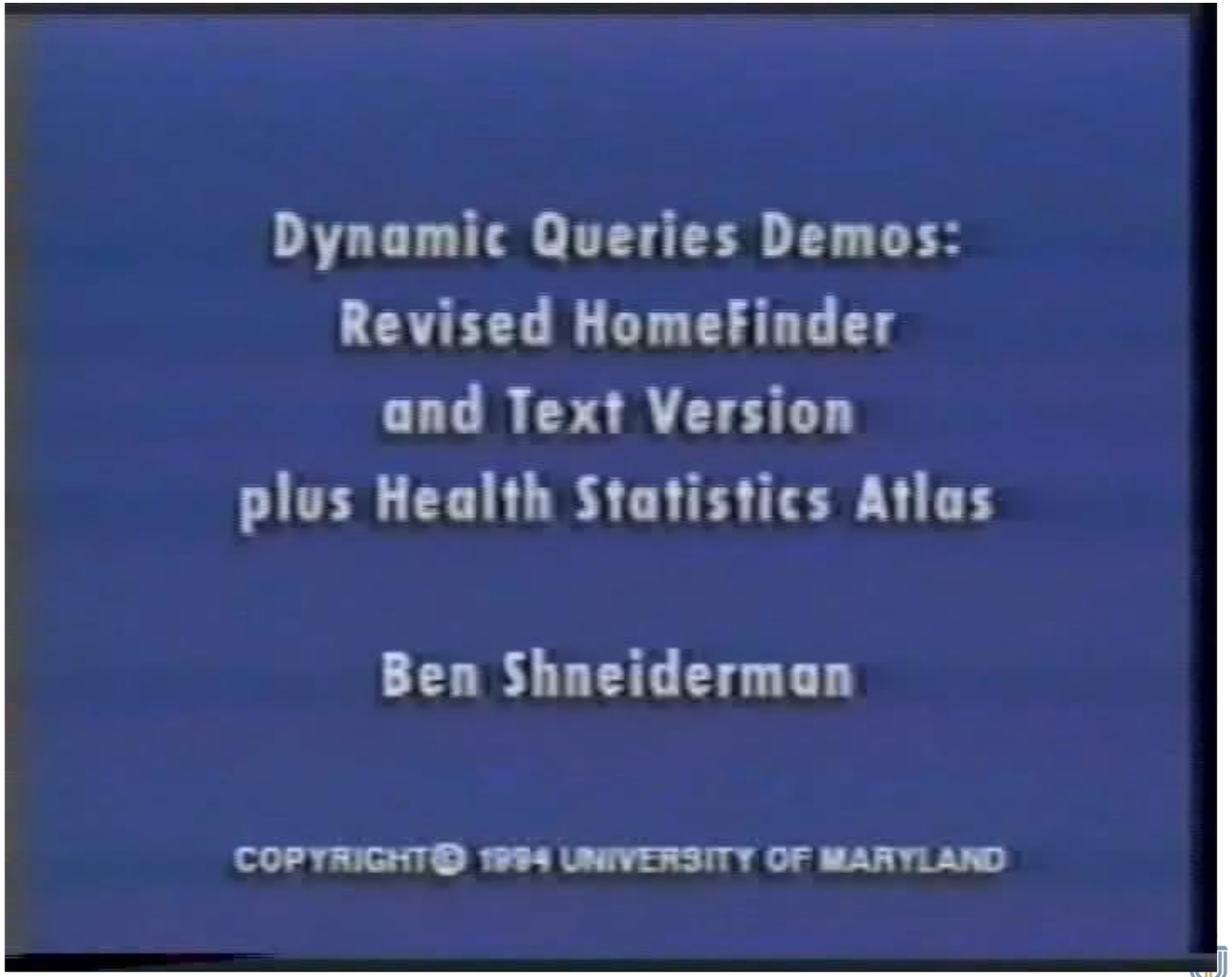
Imperfection

- Idea at heart of Dynamic Query
  - There often simply isn't one perfect response to a query
  - Want to understand a set of tradeoffs and choose some "best" compromise
  - You may learn more about your problem as you explore

Example

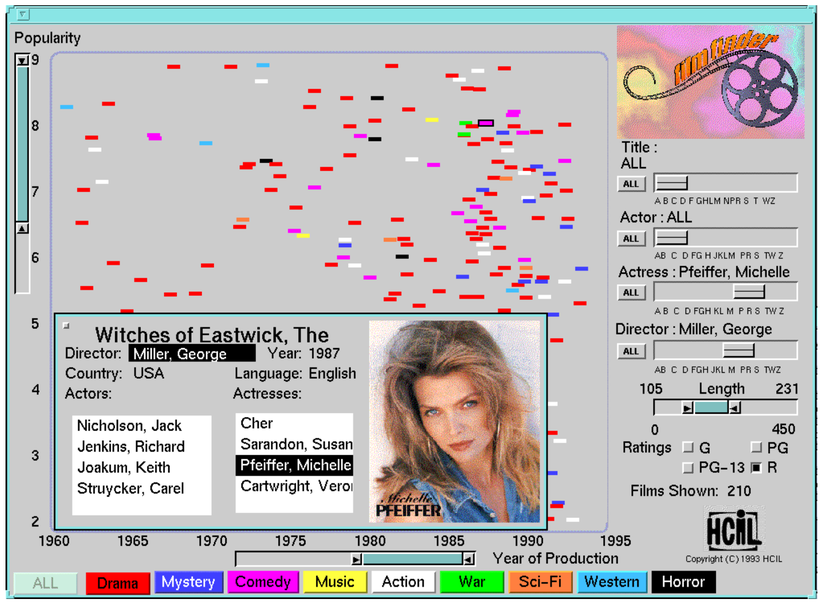
- HomeFinder
  - Univ. of Maryland

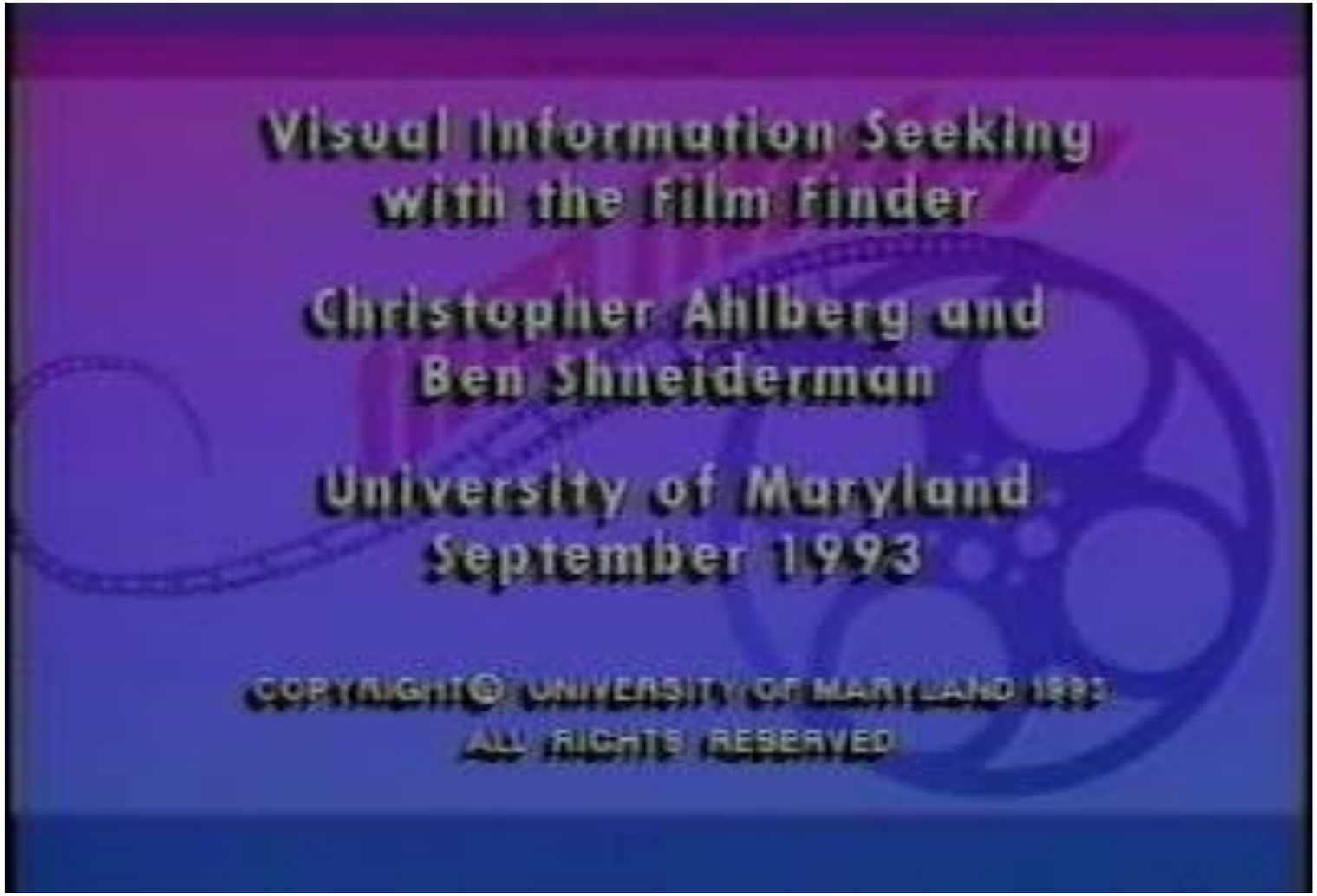




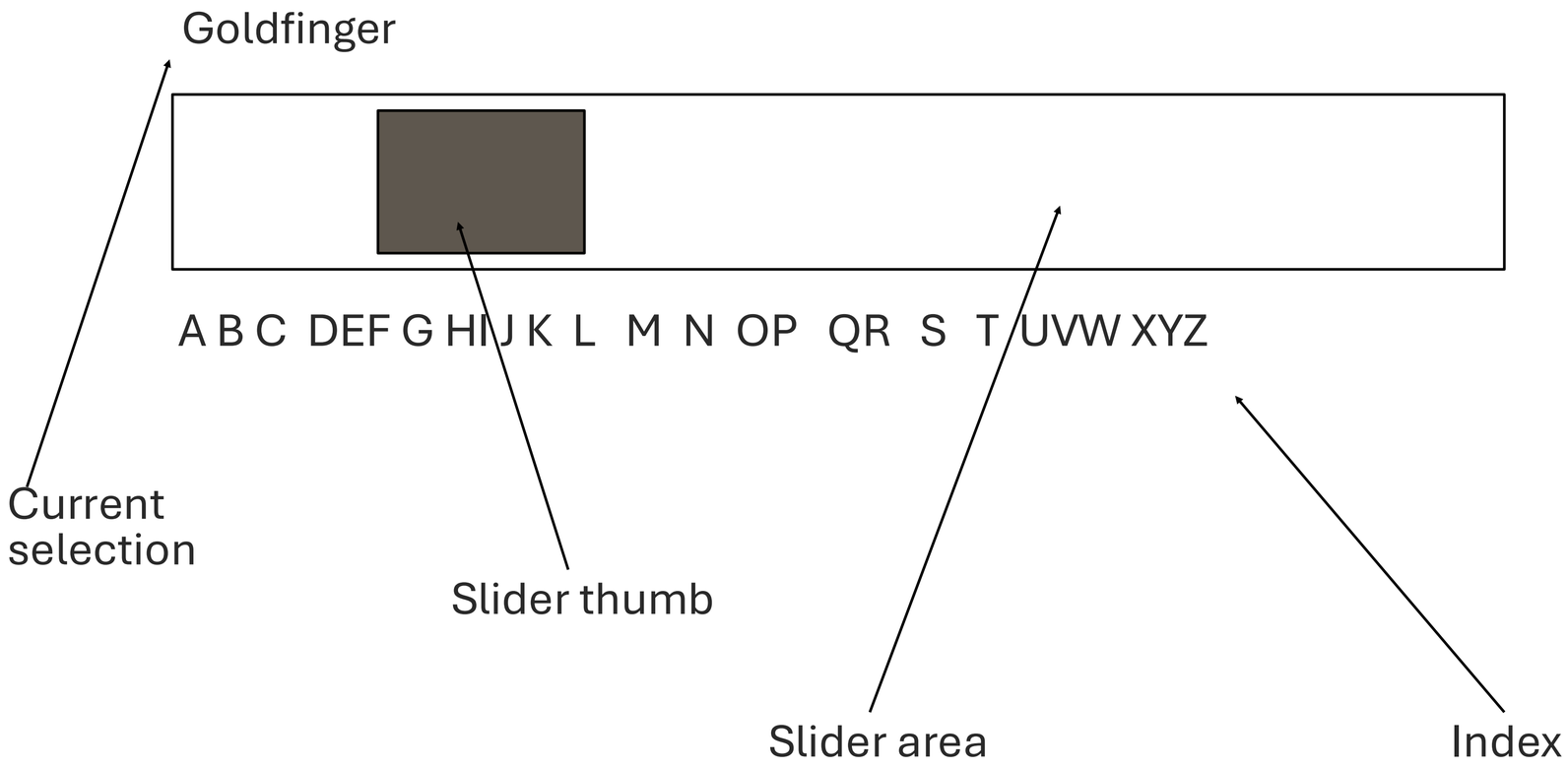
FilmFinder

C. Ahlberg
Maryland

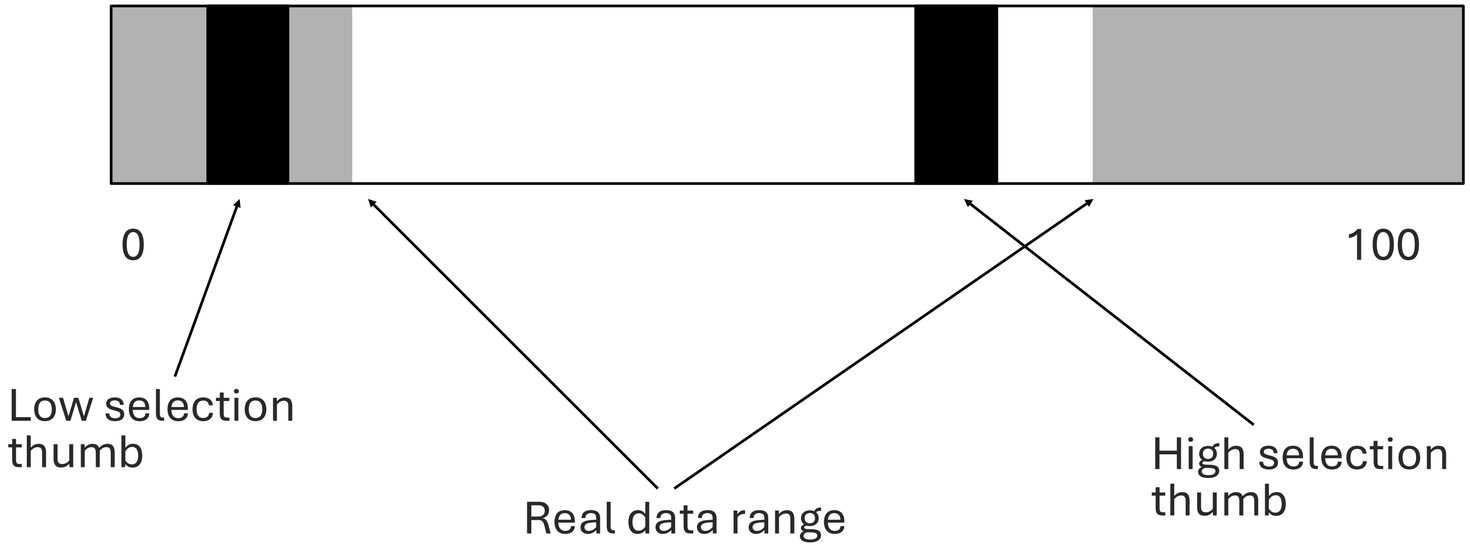




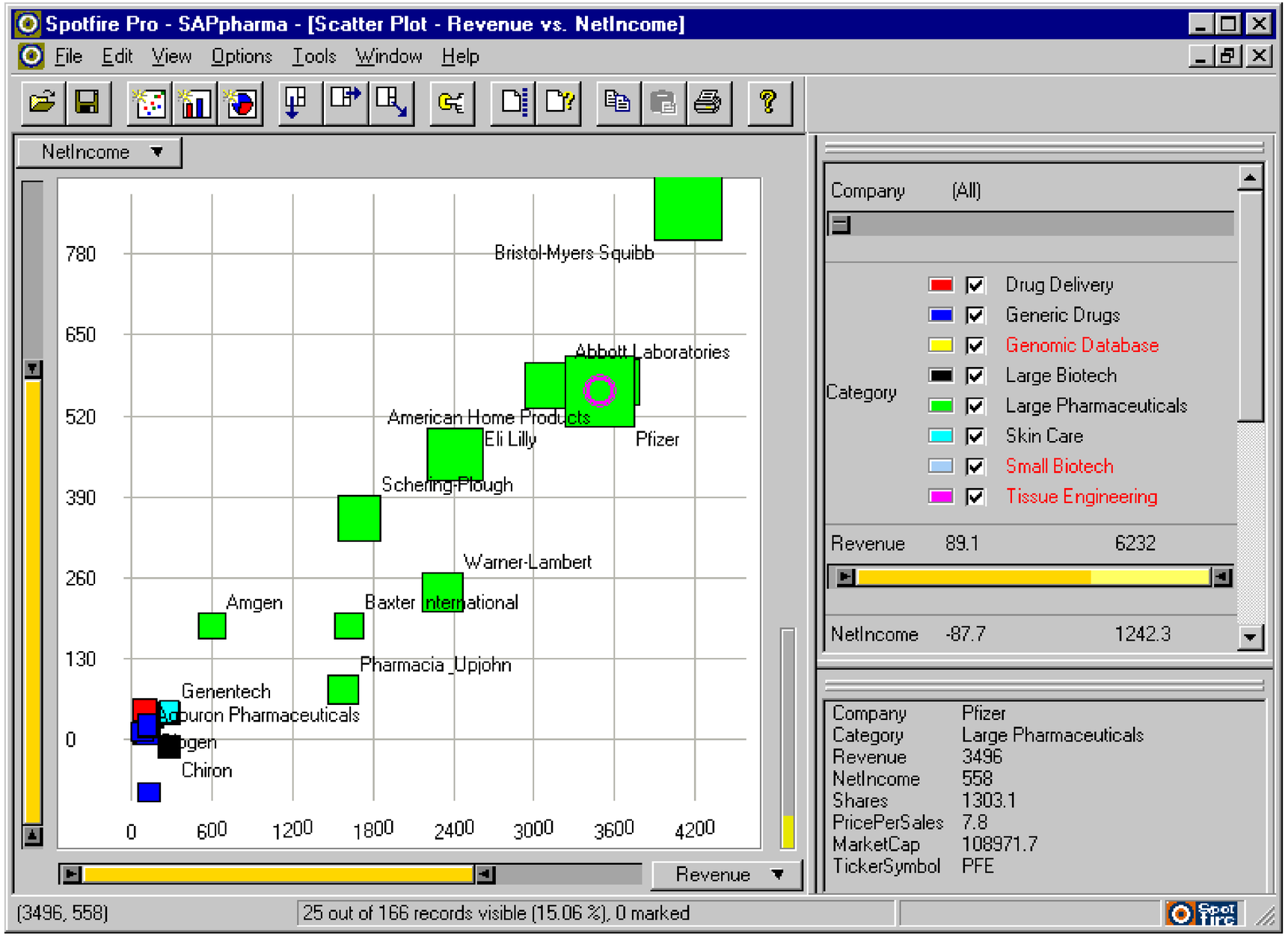
Alphaslider



Rangeslider



Spotfire


Spotfire Pro - SAPpharma - [Scatter Plot - Revenue vs. NetIncome]
File Edit View Options Tools Window Help
NetIncome
780
650
520
390
260
130
0
0 600 1200 1800 2400 3000 3600 4200
Bristol-Myers Squibb
Abbott Laboratories
American Home Products
Eli Lilly
Pfizer
Schering-Plough
Warner-Lambert
Amgen
Baxter International
Pharmacia Upjohn
Genentech
Chiron
Revenue
Company (All)
Category
Drug Delivery
Generic Drugs
Genomic Database
Large Biotech
Large Pharmaceuticals
Skin Care
Small Biotech
Tissue Engineering
Revenue 89.1 6232
NetIncome -87.7 1242.3
Company Pfizer
Category Large Pharmaceuticals
Revenue 3496
NetIncome 558
Shares 1303.1
PricePerSales 7.8
MarketCap 108971.7
TickerSymbol PFE
(3496, 558)
25 out of 166 records visible (15.06 %), 0 marked

Spotfire Features

- Starfield display
- Tight coupling
  - features to guide the user
  - rapid, incremental, reversible interactions
  - display invariants
  - continuous display
  - progressive refinement
  - details on demand

DQ Pros

- Work is faster
- Promote reversing, undo, exploration
- Very natural interaction
- Shows the data
- Easier than complex queries

DQ Cons

- Operations are fundamentally conjunctive
- Controls are global in scope
  - They affect everything

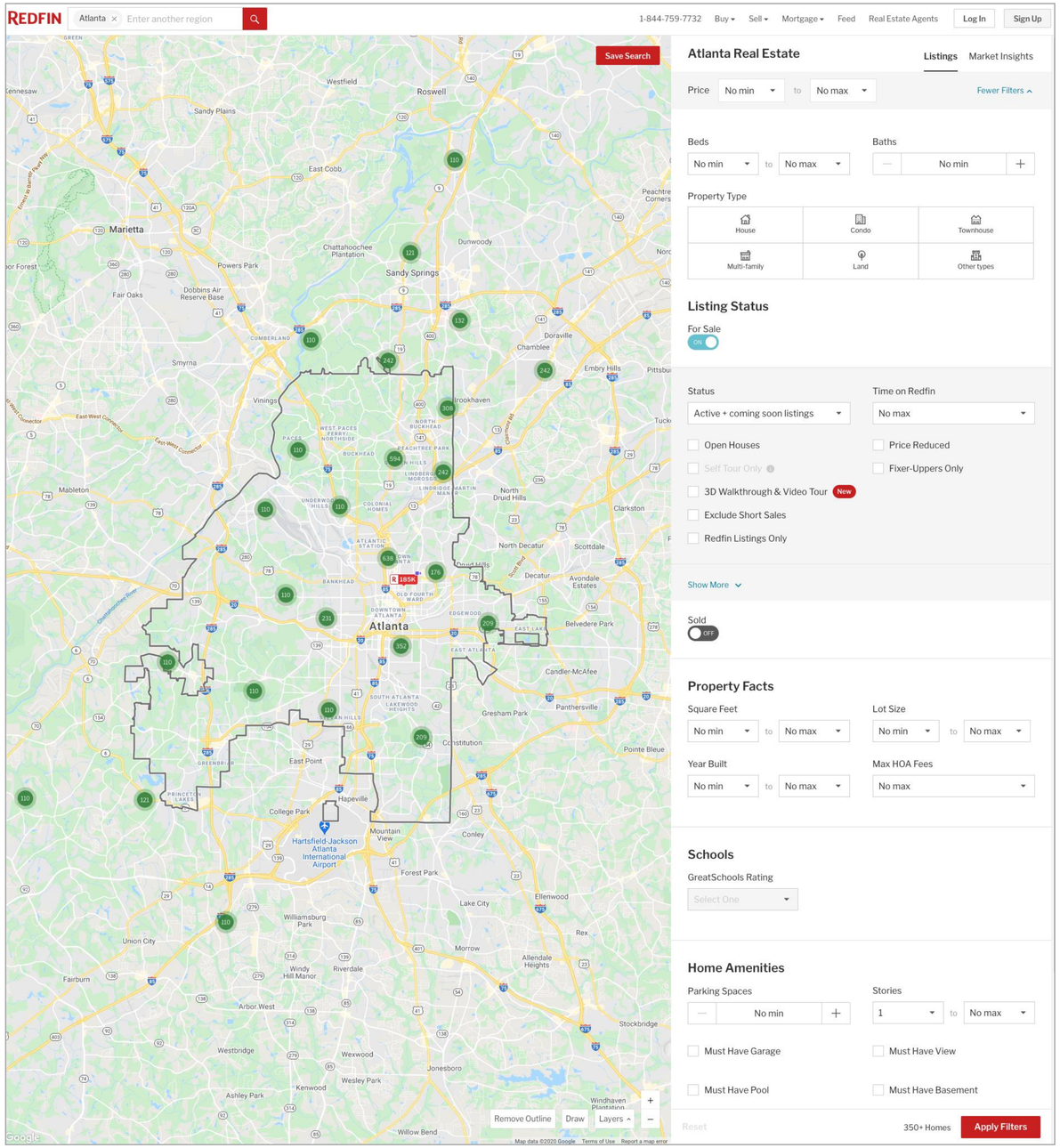
- Controls must be fixed in advance
  - you have to program what each slider controls

“Interaction Junk”
Endert and North, BELIV 2012

DQ Cons

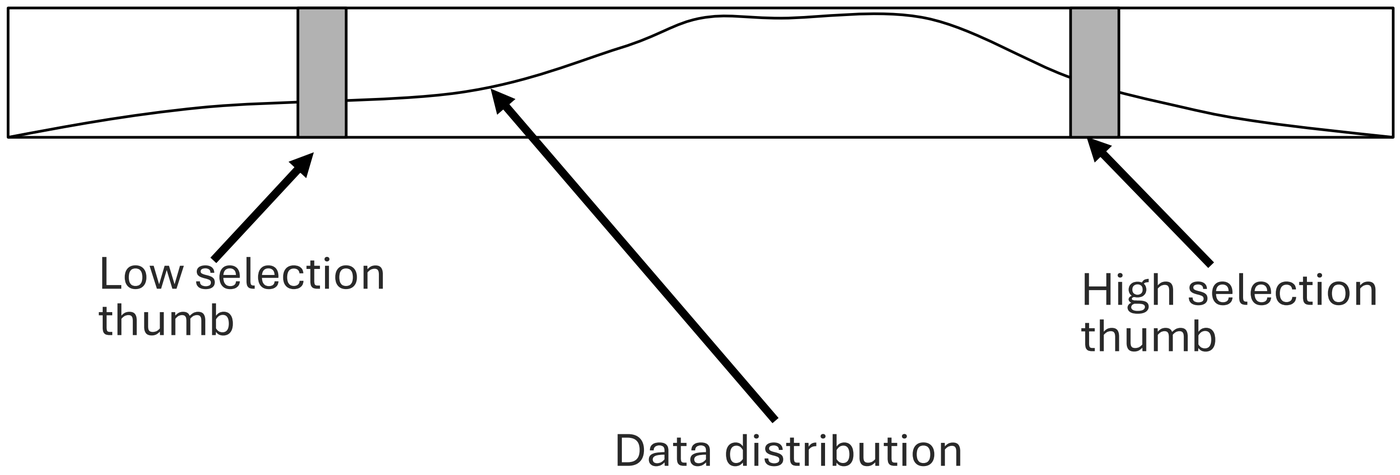
- Controls take space!
  - How much in Spotfire?

- Put data in controls?
- Put controls in the vis?



Data Visualization Sliders

- Data in the controls…

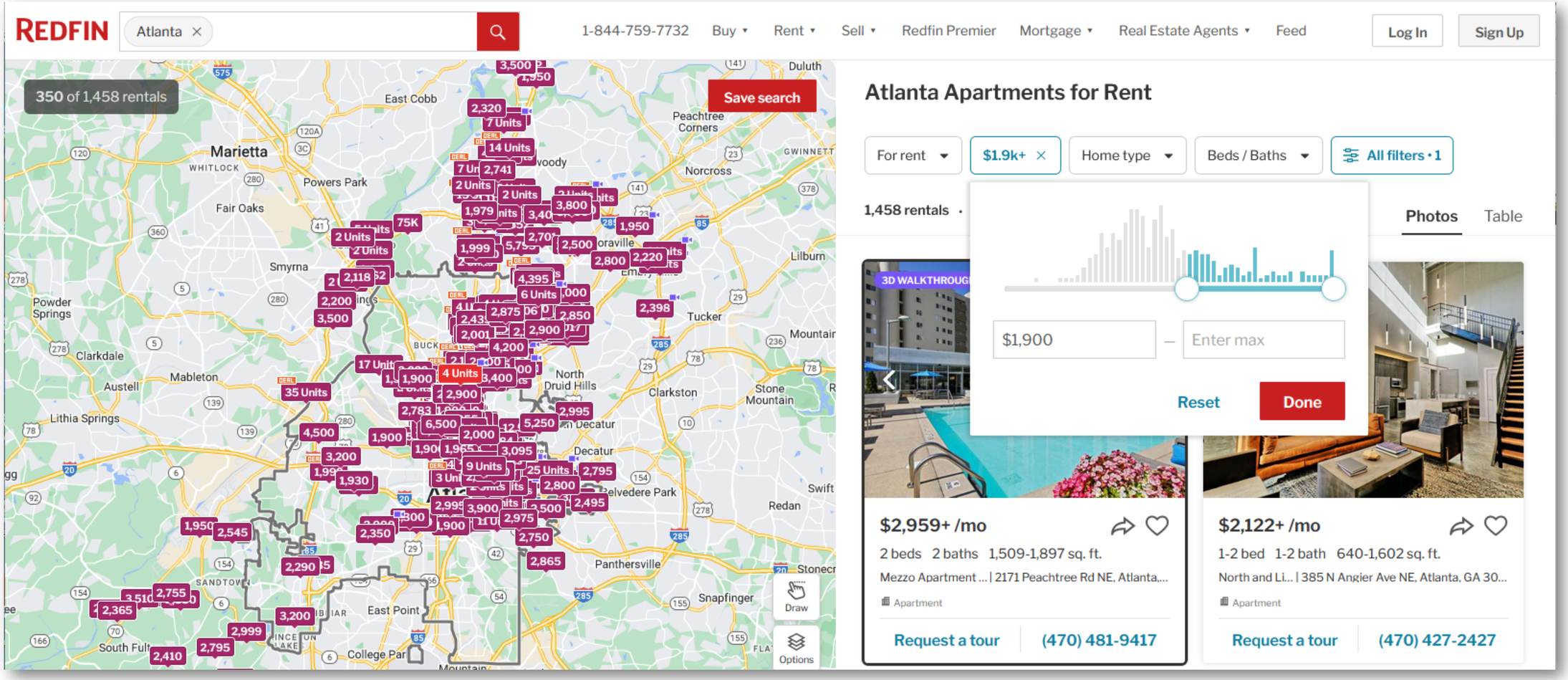


Eick
UIST '94

Apt Prices

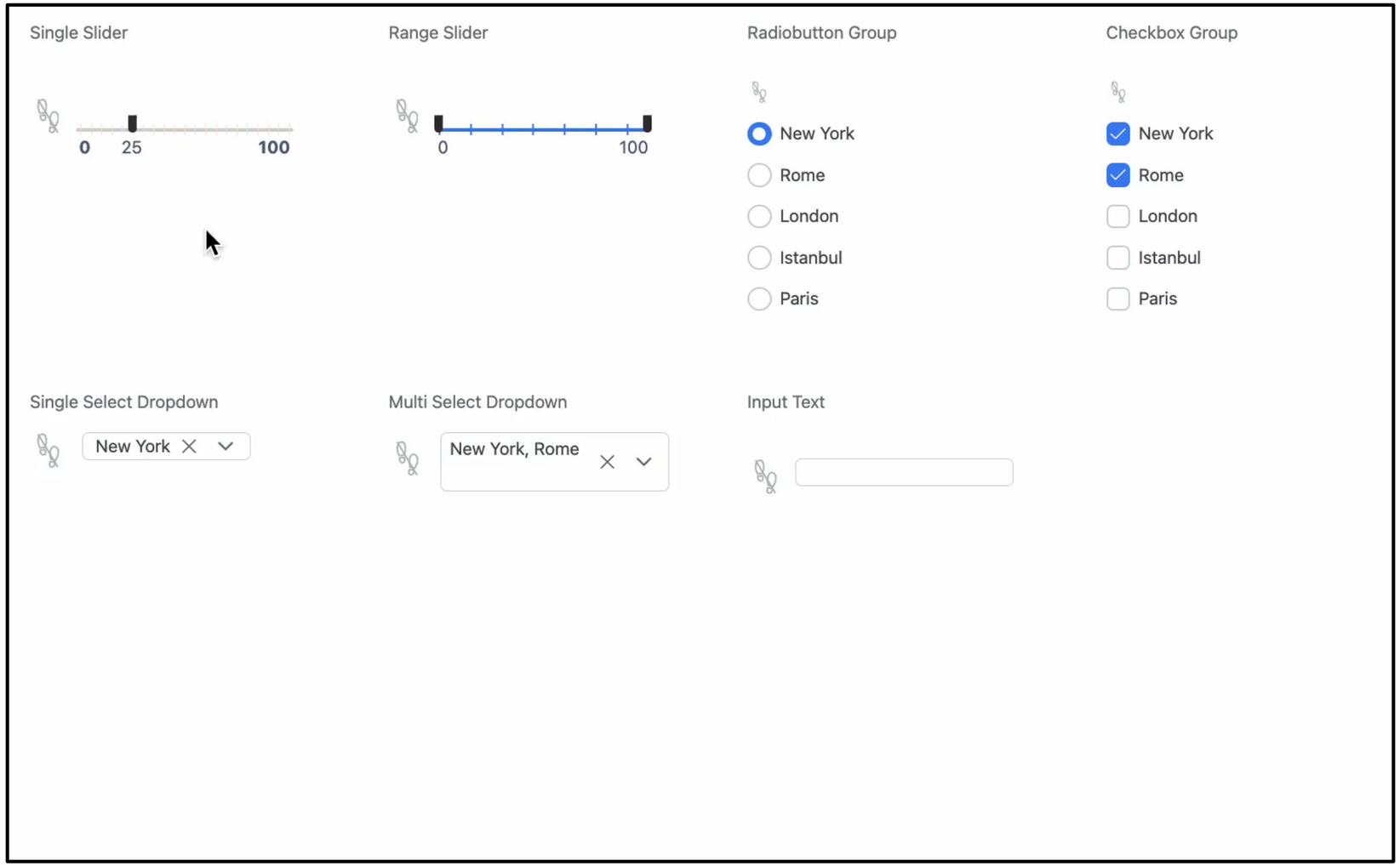
https://www.redfin.com

Doesn't update as you move the slider though



ProvenanceWidgets

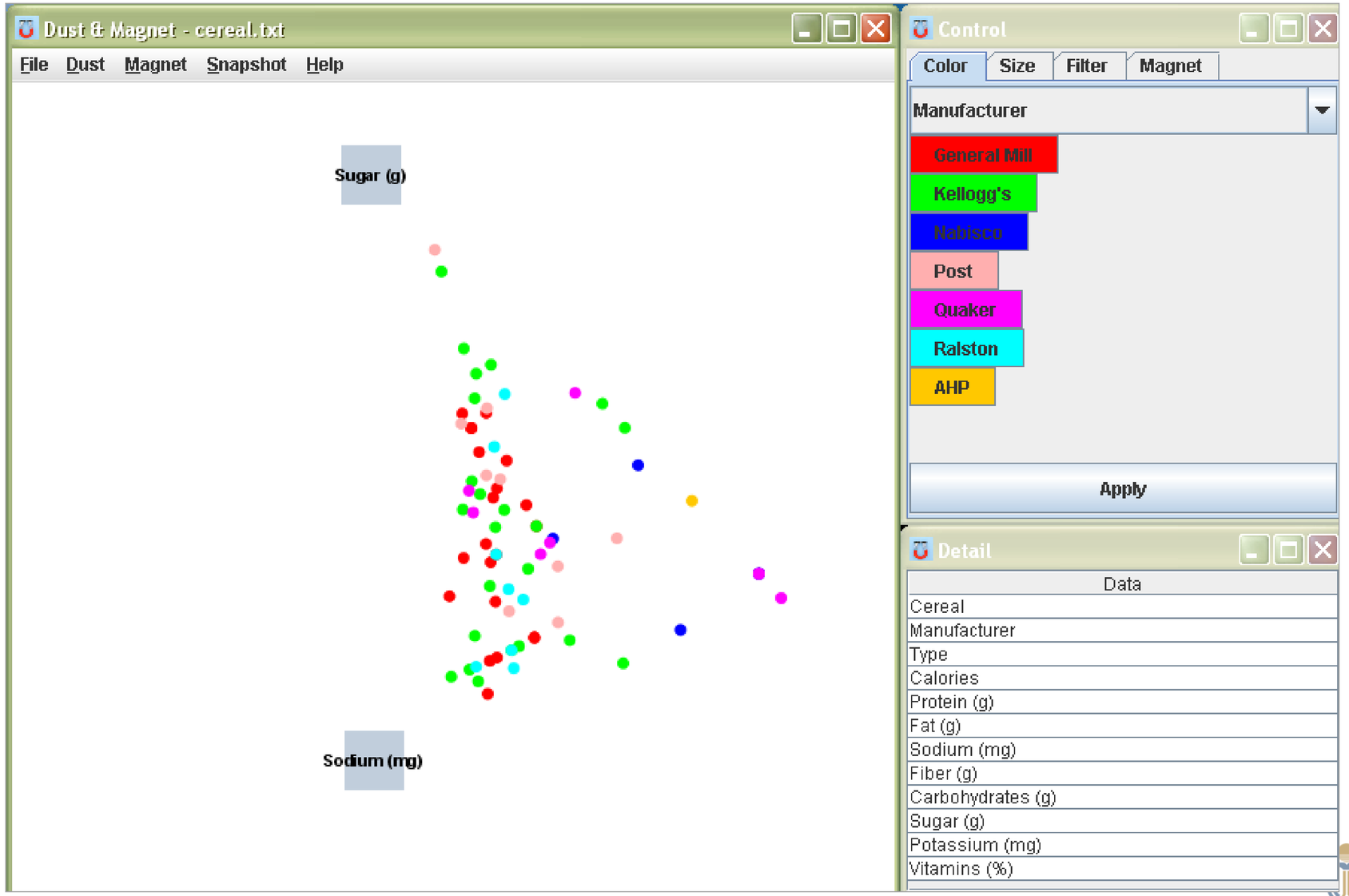
A Library of UI Control Elements to Track and Dynamically Overlay Analytic Provenance



Narechania
VIS 2024

Dust & Magnet

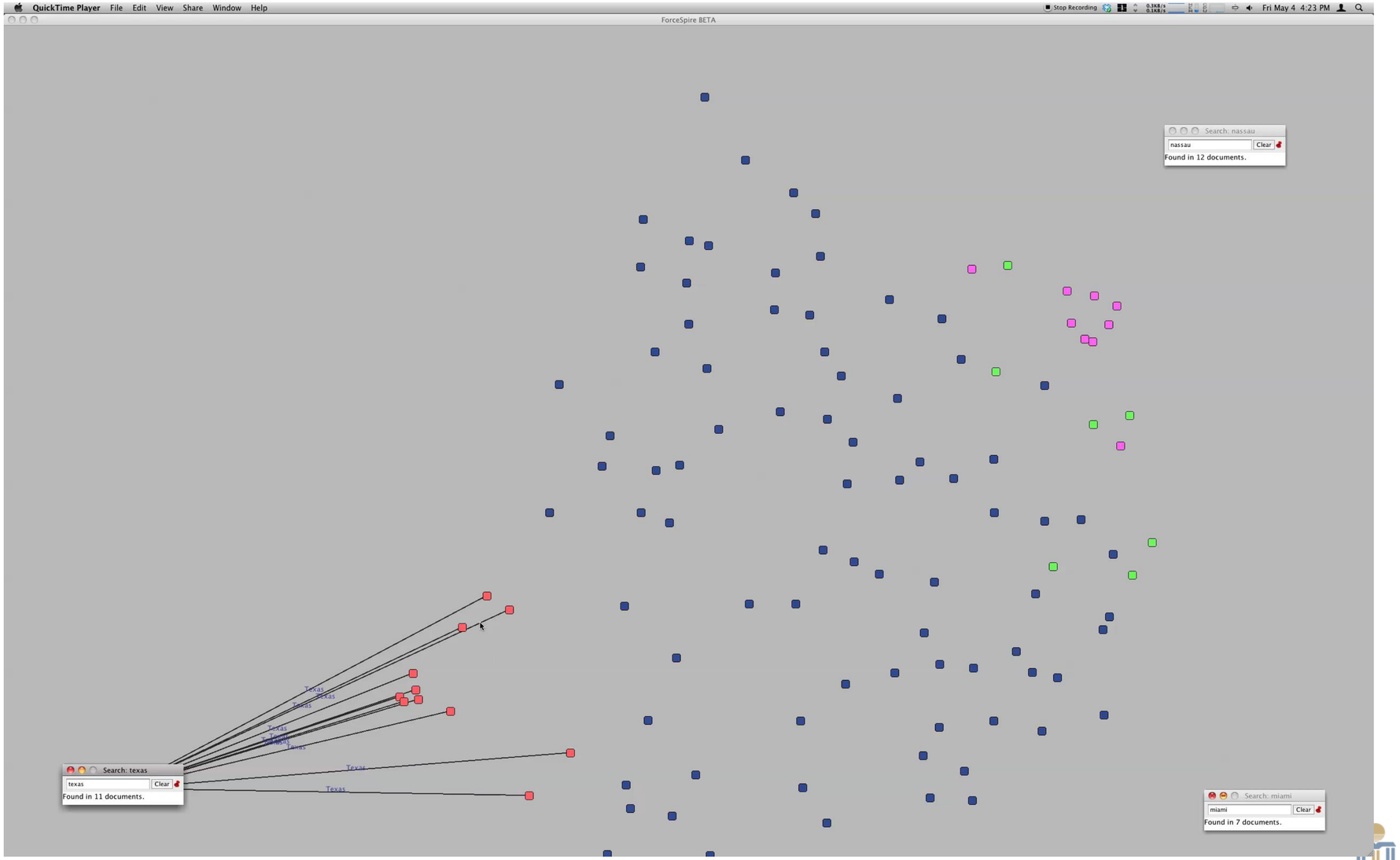
- Control in the visualization…



Yi et al
Information Visualization ‘05

Semantic Interaction

- Control in the visualization…

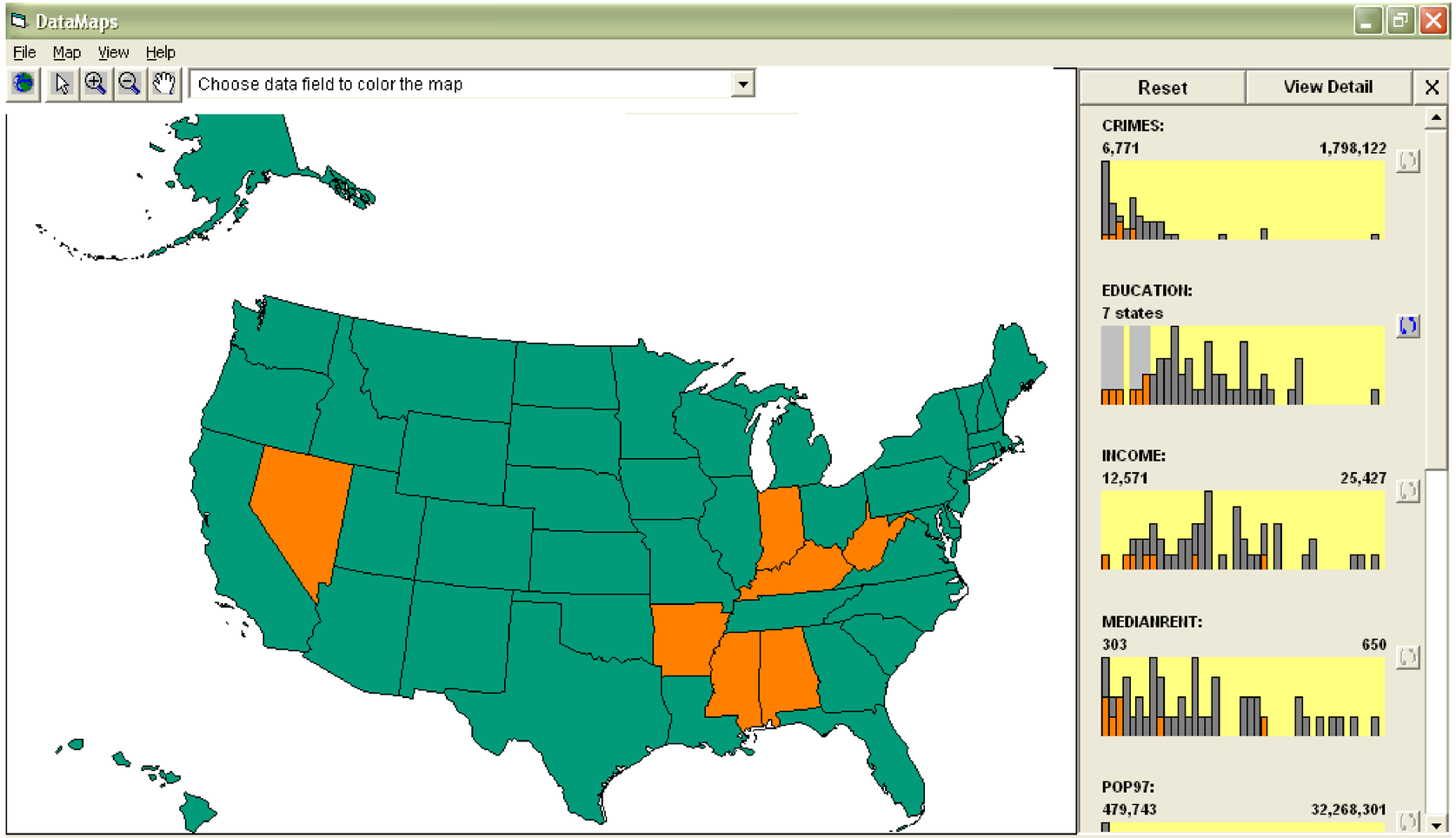


Endert

Brushing Histograms

- Special case of brushing
- Data values represented in histograms that can be clicked on and selected (controls region)
- When items selected there, the corresponding item(s) are highlighted in main view windows

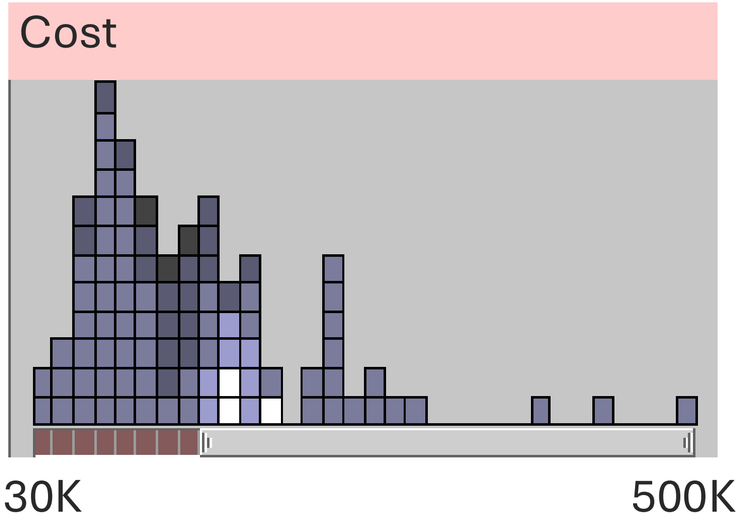
DataMaps

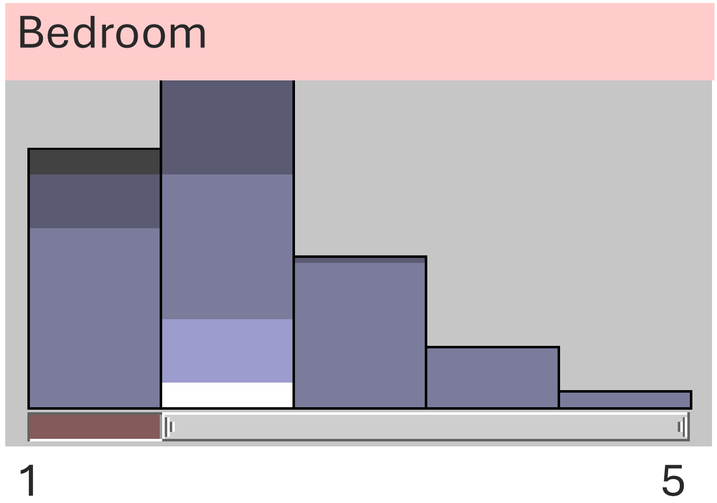


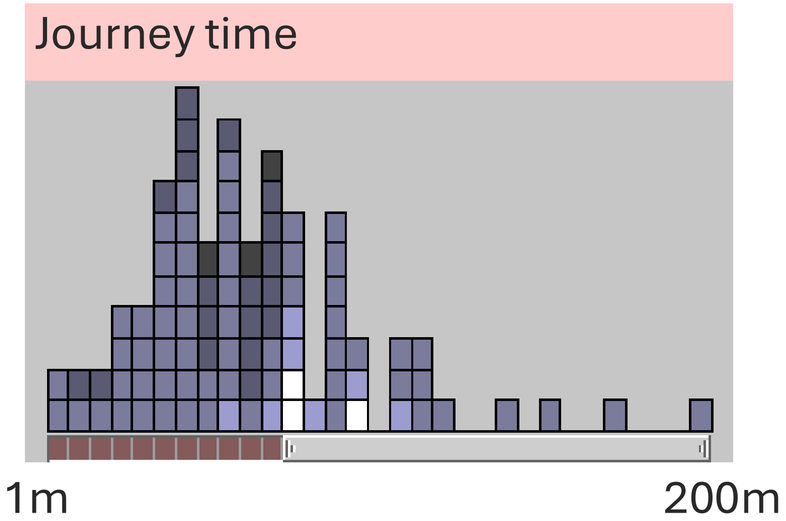
State	CRIMES	EDUCATION	INCOME	MEDIANRENT	POP97
Nevada	99,288	15.3	20,254	509	1,676,809
West Virginia	44,879	12.3	13,967	303	1,815,787
Indiana	201,604	15.6	16,816	374	5,864,108
Kentucky	121,047	13.6	14,747	319	3,908,124
Arkansas	116,206	13.3	13,784	328	2,522,819
Mississippi	79,958	14.7	12,571	309	2,730,501
Alabama	203,382	15.7	14,903	325	4,319,154

Spence & Tweedie
Inter w Computers ‘98

Attribute Explorer







Attribute histogram
All objects on all attribute scales
Interaction with attributes limits
Brushing across views
Color-encoded sensitivity

Magic Lenses

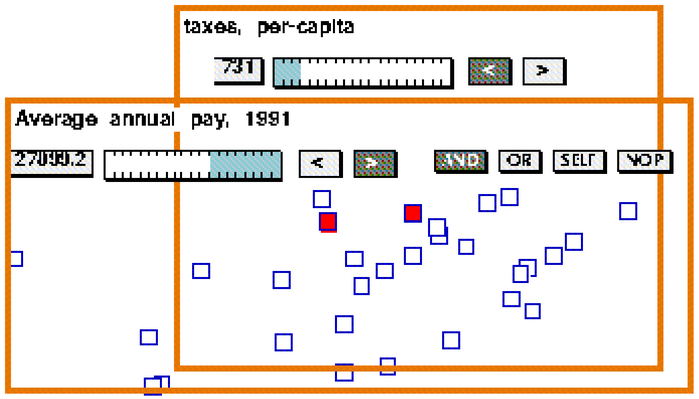
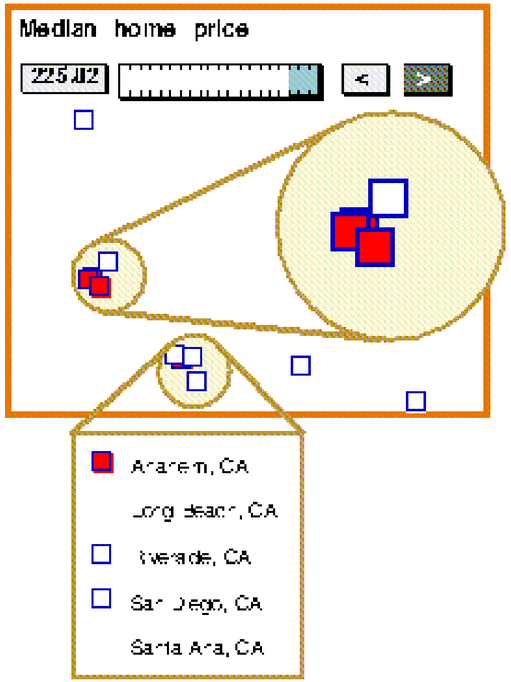


Figure 1(a) High salaries AND low taxes.



Fishkin & Stone
CHI '95

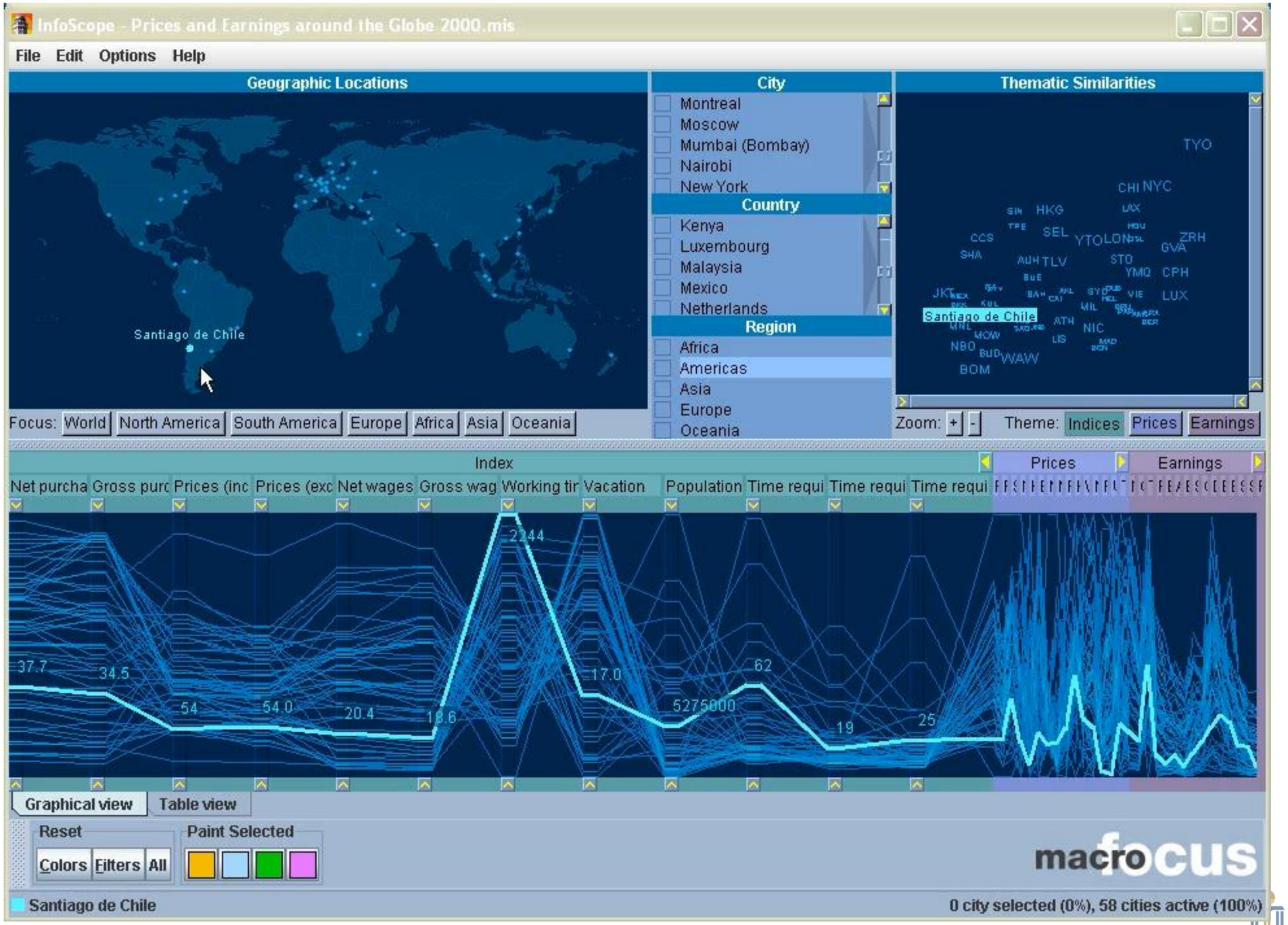
Magic Lenses

Brushing & Linking

- Applies when you have multiple views of the same data
- Selecting or highlighting a case in one view generates highlighting the case in the other views
- Very common technique in InfoVis

Example

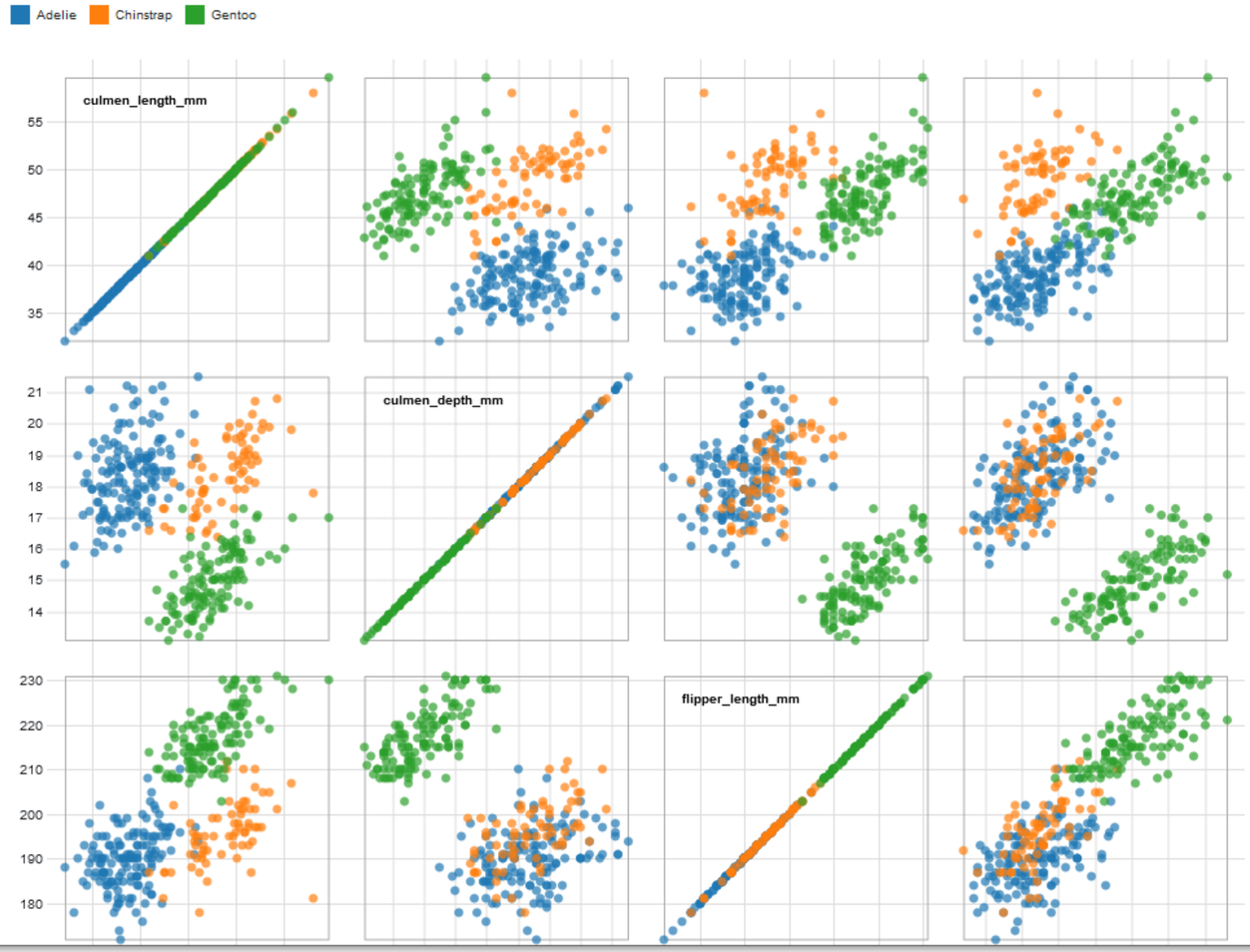
InfoScope



https://observablehq.com/@d3/brushable-scatterplot-matrix

Brushable Scatterplot Matrix

This scatterplot matrix allows brushing to select data points in one cell, and highlight them across all other cells.



Key Points so far

- Interaction facilitates a dialog between the user and the visualization system
- Multiple views amplify importance of interaction
- Interaction often helps when you just can’t show everything you want

Review

- Define interaction - what is it?
- Talk about taxonomies for interaction
- Discuss examples for types of interaction

Next Time

- Interaction (contd)